Textbooks and Learning Resources

- Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg, Andrzej M. Goscinski.
- Cloud Computing: Principles, Systems and Applications by Nikos Antonopoulos, Lee Gillam
- CompTIA Cloud+ Guide to Cloud Computing, 2nd Edition, 2023, Jill West, ISBN-13: 978-0-357-88345-7
- Class Lecture Documents.

Course contents

- Introduction to Cloud Computing
- Cloud Infrastructure
- Virtualization of Physical Systems
- Migration to the Cloud
- Cloud Connectivity and Troubleshooting
- Securing Cloud Resources
- Identity and Access Management
- Cloud Storage
- Managing Cloud Capacity and Performance
- Cloud Automation

Introduction

- Definition of Cloud Computing
- Evolution process of Cloud Computing
- Cloud computing services
- Benefits, Challenges and Risks of cloud computing
- Enabling technologies and deployment models
- Cloud security issues
- Economic models and SLAs
- Network and Protocol stacks
- Popular cloud stacks and use cases

Cloud Computing

- Cloud computing is a technology to transform products into services over the internet.
- It allows users to access and use computing resources such as storage, computing power, networking, databases etc.
- On-demand delivery of computer system services.

Cloud Computing



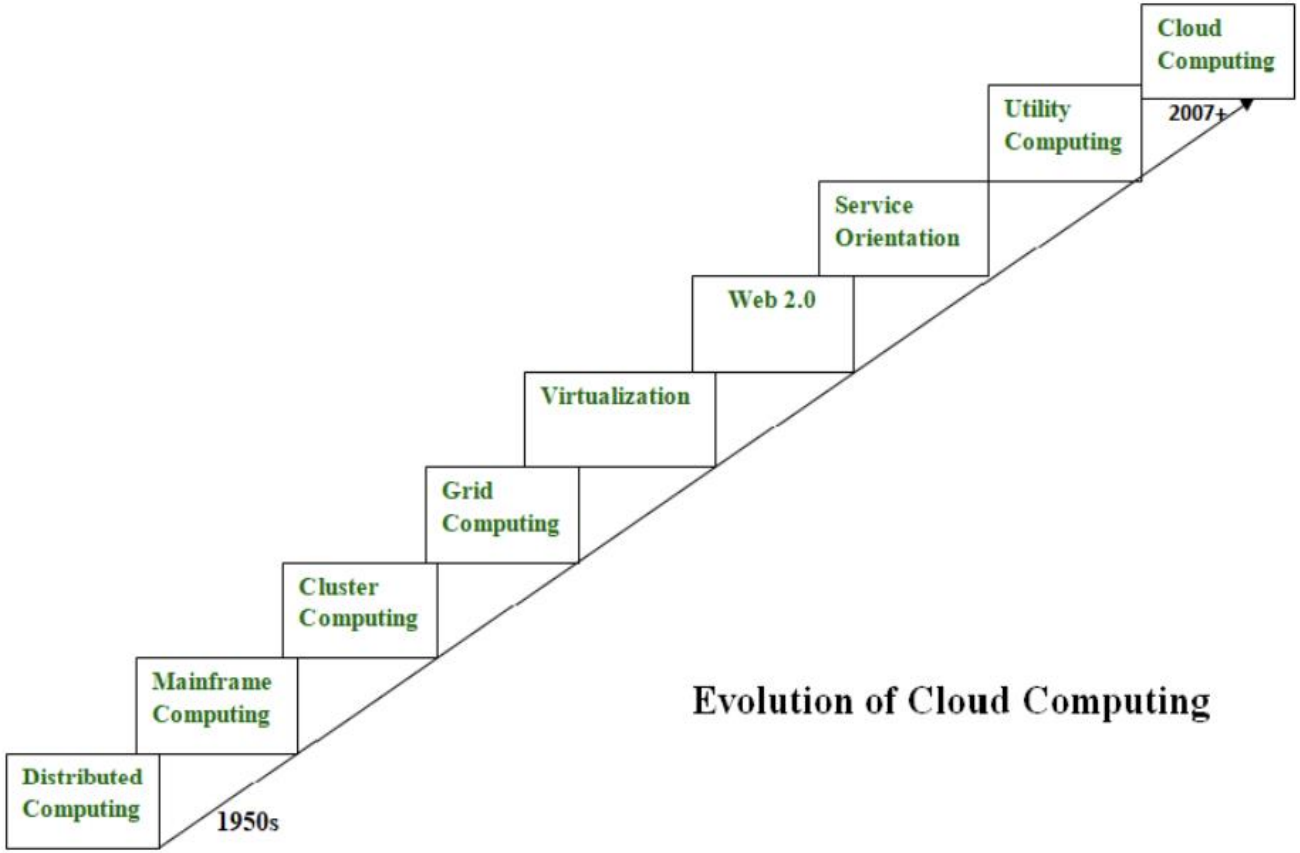
Evolution

- First introduced in 1950s and it was evolved from the term distributed computing.

-Mainframe computing in 1951 is capable of handling large data operations.

-Cluster computing in 1980s.

-Grid computing in 1990s.



Cloud Computing

Advantages of cloud computing:

- ❏ Cost saving
- ❏ Data redundancy and replication
- ❏ Scalability
- ❏ Flexibility
- ❏ Reliability
- ❏ High accessibility
- ❏ Malware protection

Cloud Computing

Disadvantages:

- Internet dependency
- Limitations on control
- Issues in security and privacy
- Data breaches

Cloud Computing Services

- Infrastructure-as-a-service (IaaS)
- Platform-as-a-service (PaaS)
- Software-as-a-service (SaaS)

IaaS

- Delivery of computing infrastructure as on demand service.
- Users can purchase servers, software data centers, network equipments based on need.
- Additionally, users are able to rent those resources through a fully outsourced on demand service.
- It allows dynamic scaling and the resources are distributed.
- Generally includes multiple users on a single piece of hardware.

- IAAS is like renting virtual computers and storage space in the cloud.
- You have control over the operating systems, applications, and development frameworks.
- Scaling resources up or down is easy based on your needs.

Examples

- Amazon web services
- Microsoft Azure
- Google Compute Cloud

PaaS

- PAAS is like a toolkit for developers to build and deploy applications without worrying about infrastructure.
- Provides pre-built tools, libraries, and development environments.
- Developers focus on building and managing applications, while the provider handles infrastructure management.
- It speeds up the development process and allows for easy collaboration among developers.

PaaS

Examples

- AWS Lambda
- Google cloud
- Google App engine

SaaS

- Applications are ready to use, and updates and maintenance are handled by the provider.
- You access the software through a web browser or app, usually paying a subscription fee.
- It’s convenient and requires minimal technical expertise, ideal for non-technical users.

SaaS

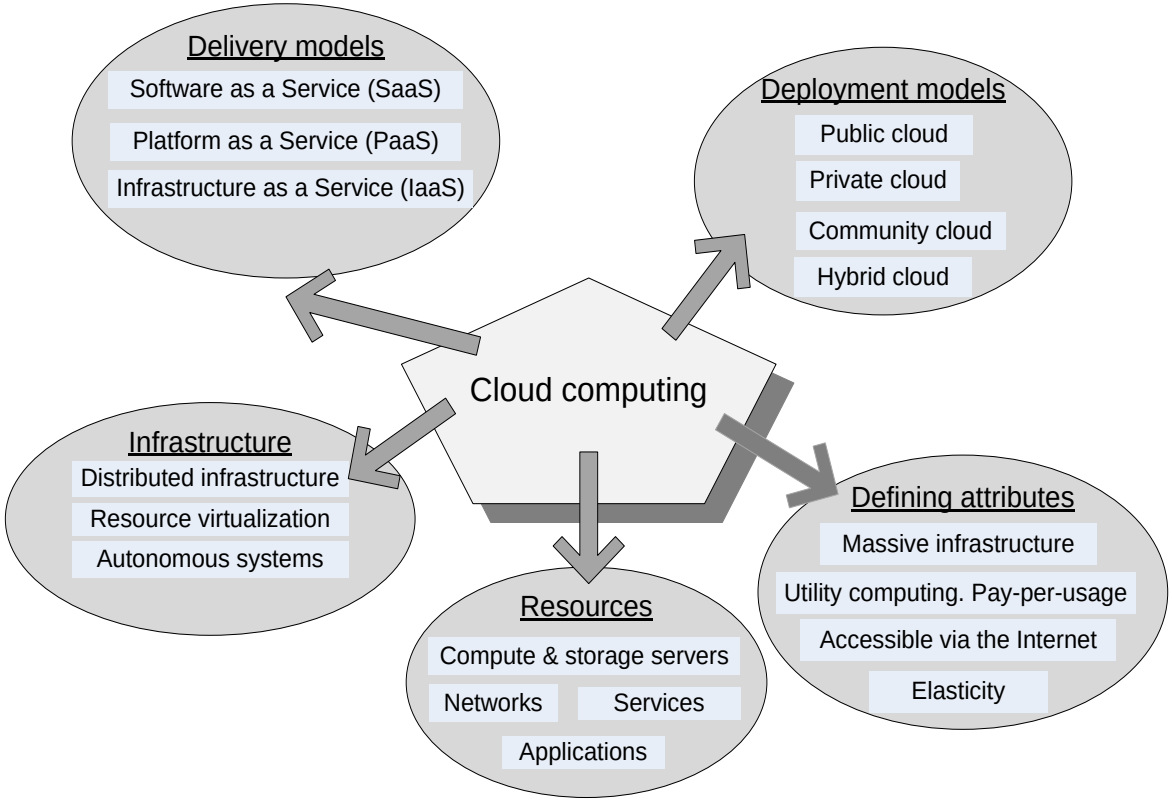
Examples

- Salesforce
- Google Workspace apps
- Microsoft 365
- Trello
- Zoom
- Slack
- Adobe Creative Cloud

Cloud Computing Models

- Public cloud
- Private cloud
- Hybrid cloud

Cloud Computing



Types of clouds

1. **Public Cloud** - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services e.g. AWS, Google, Microsoft.

2. **Private Cloud** – the infrastructure is operated solely for an organization e.g. Dell, IBM, Oracle.

1. **Hybrid Cloud** - composition of two or more Clouds (public, private, or community) as unique entities but bound by a standardized technology that enables data and application portability.

2. **Other types: e.g., Community/Federated Cloud** - the infrastructure is shared by several organizations and supports a community that has shared concerns.

Why cloud computing is (could) be successful when other paradigms have failed?

- It is in a better position to exploit recent advances in software, networking, storage, and processor technologies promoted by the same companies who provide Cloud services.
- Economical reasons: It is used for enterprise computing; its adoption by industrial organizations, financial institutions, government, and so on has a huge impact on the economy.

Infrastructures Management reasons:

- A single Cloud consists of a mostly homogeneous (now more heterogeneous) set of hardware and software resources.
- The resources are in a single administrative domain (AD). Security, resource management, fault-tolerance, and quality of service are less challenging than in a heterogeneous environment with resources in multiple ADs.

Challenges for cloud computing

1. Availability of service: what happens when the service provider cannot deliver?

2. Data confidentiality and auditability, a serious problem.

3. Diversity of services, data organization, user interfaces available at different service providers limit user mobility; once a customer is hooked to one provider it is hard to move to another.

4. Data transfer bottleneck; many applications are data-intensive.

Challenges for cloud computing

5. Performance unpredictability, one of the consequences of resource sharing.

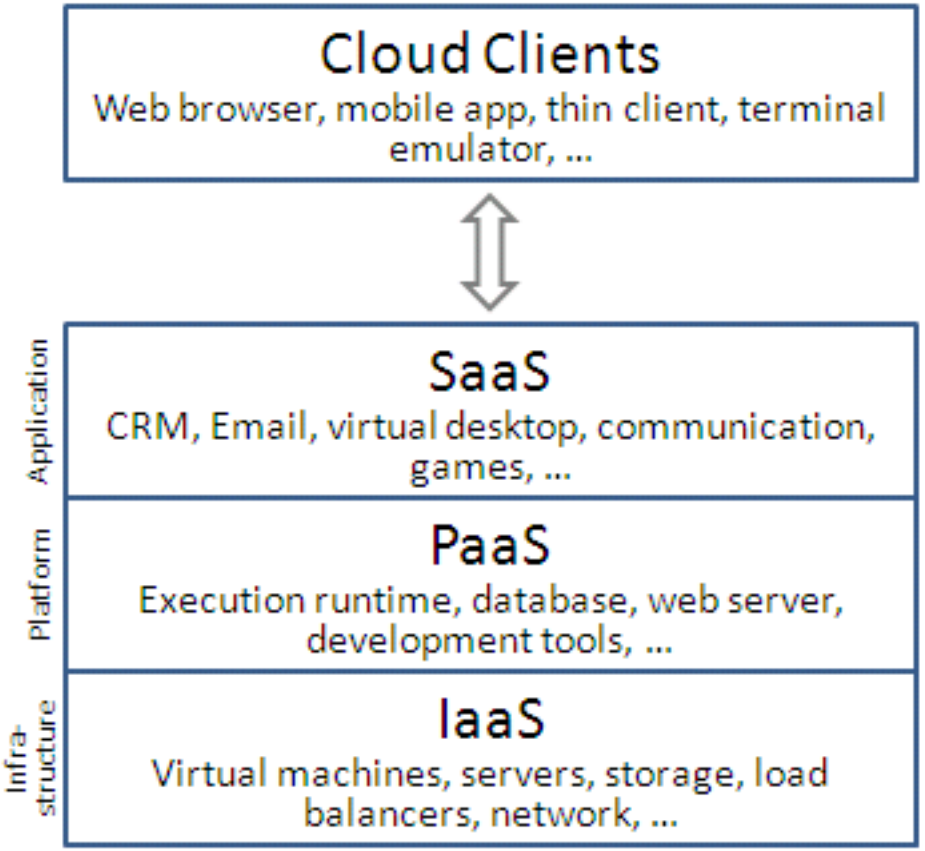
How to use resource virtualization and performance isolation for QoS guarantees?
How to support elasticity, the ability to scale up and down quickly?

6. Resource management: It is a big challenge to manage different workloads running on large data centers. Are self-organization and self-management the solution?

7. Security and confidentiality: major concern for sensitive applications, e.g., healthcare applications.

Cloud Delivery Models

1. **Software as a Service (SaaS)** (high level)
2. **Platform as a Service (PaaS)**
3. **Infrastructure as a Service (IaaS)** (low level)



The Three delivery models of Cloud Computing

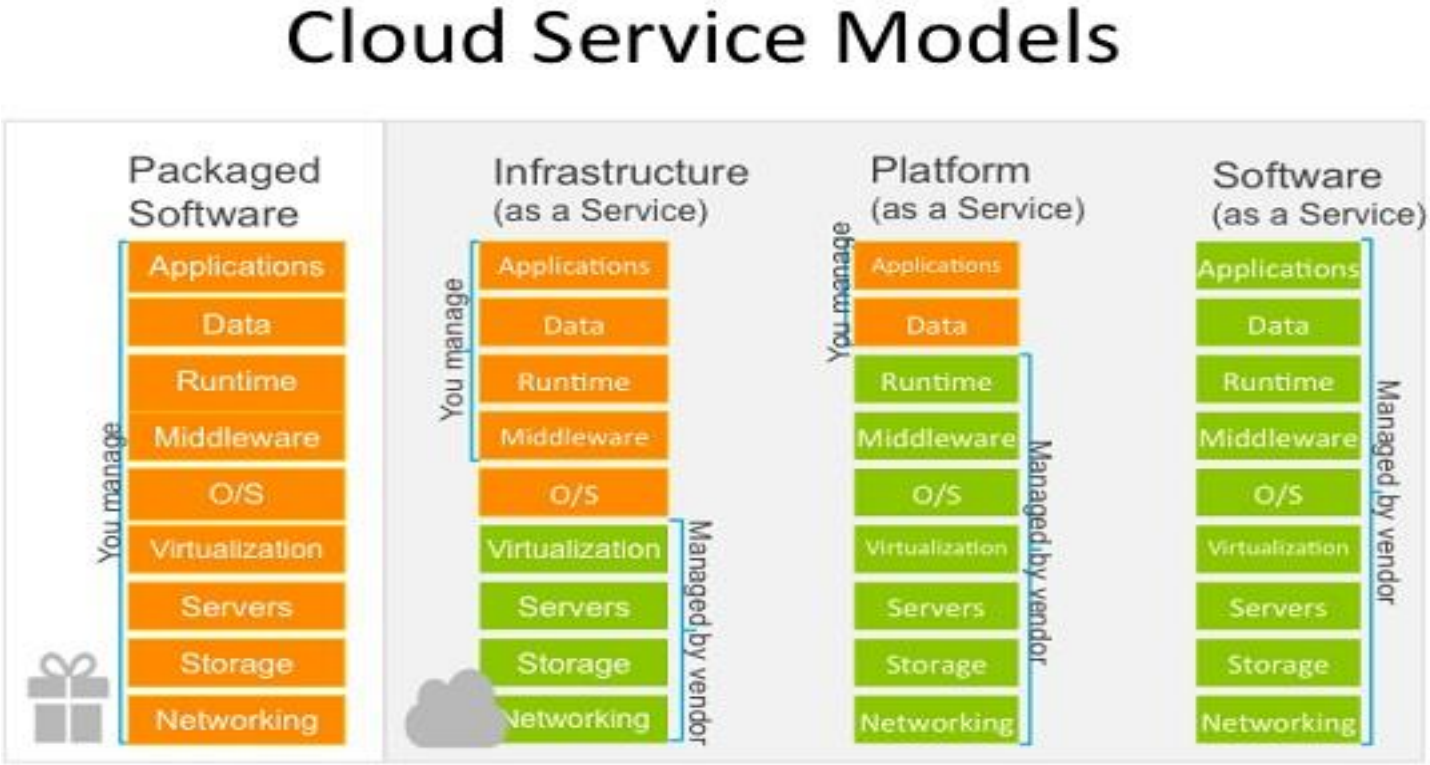


Figure 1. **Source: Microsoft Azure**

Cloud activities

Service management and provisioning including:

- ❑ Virtualization.
- ❑ Service provisioning.
- ❑ Call center.
- ❑ Operations management.
- ❑ Systems management.
- ❑ QoS management.
- ❑ Billing and accounting, asset management.
- ❑ SLA management.
- ❑ Technical support and backups.

Cloud activities (cont’d)

Security management including:

- ID and authentication.
- Certification and accreditation.
- Intrusion prevention.
- Intrusion detection.
- Virus protection.
- Cryptography.
- Physical security, incident response.
- Access control, audit and trails, and firewalls.

Cloud activities (cont’d)

Customer services such as:

- Customer assistance and on-line help.
- Subscriptions.
- Business intelligence.
- Reporting.
- Customer preferences.
- Personalization.
- Integration services including:
- Data management.
- Development.

Ethical issues

Paradigm shift with implications on computing ethics:

- The control is relinquished to third party services.
- Data is stored on multiple sites administered by several organizations.
- Multiple services interoperate across the network.

Implications:

- Unauthorized access.
- Data corruption.
- Infrastructure failure, and service unavailability.

De-perimeterization

- Systems can span the boundaries of multiple organizations and cross the security borders.

- The complex structure of Cloud services can make it difficult to determine who is responsible in case something undesirable happens.

- Identity fraud and theft are made possible by the unauthorised access to personal data in circulation and by new forms of dissemination through social networks and they could also pose a danger to Cloud Computing.

Privacy issues

- Cloud service providers have already collected petabytes of sensitive personal information stored in data centers around the world. The acceptance of Cloud Computing therefore will be determined by privacy issues addressed by these companies and the countries where the data centers are located.

- Privacy is affected by cultural differences; some cultures favour privacy, others emphasize community. This leads to an ambivalent attitude towards privacy in the Internet which is a global system.

Cloud Vulnerabilities

- Clouds are affected by malicious attacks and failures of the infrastructure, e.g., power failures.

Such events can affect the Internet domain name servers and prevent access to a

Cloud or can directly affect the Clouds:

- in 2004 an attack at Akamai caused a domain name outage and a major blackout that affected Google, Yahoo, and other sites.
- in 2009, Google was the target of a denial of service attack which took down Google News and Gmail for several days;
- in 2012 lightning caused a prolonged down time at Amazon.

Basics : Parallel Computing

***"Parallel computing** is a form of computation in which many calculations are carried out simultaneously, operating on the principles that large problems can often be divided into smaller ones, which are then solved concurrently (in parallel)."*

Hardware and software systems allow us to:

- Solve problems demanding resources not available on a single system.
- Reduce the time required to obtain a solution.

Parallel Computing – Amdahl's Law

The speedup *S* measures the effectiveness of parallelization:

$$S(N) = T(1) / T(N)$$

T(1) → the execution time of the sequential computation.
T(N) → the execution time when N parallel computations are executed

Amdahl's Law: if α is the fraction of running time a sequential program spends on non-parallelizable segments of the computation then:

$$S \cong 1/\alpha$$

This is a theoretical upper bound on the best speedup we can get from parallelising a certain program.

Basics : Distributed Systems

Collection of autonomous computers, connected through a network and distribution software (often) called middleware which enables computers to coordinate their activities and to share system resources for a common goal.
Characteristics:

1. The users perceive the system as a single, integrated computing facility.
2. The components are autonomous.
3. Scheduling and other resource management and security policies are implemented by each system.
4. There are multiple points of control and multiple points of failure.
5. The resources may not be accessible at all times.
6. Can be scaled by adding additional resources.
7. Can be designed to maintain availability even at low levels of hardware/software/network reliability.

Virtualization

Virtualization

- Three fundamental abstractions are necessary to describe the operation of a computing systems:
  (1) interpreters/processors, (2) memory, (3) communications links

As the scale of a system and the size of its users grows, it becomes very challenging to manage its recourses (see three points above)

Resource management issues:

- provision for peak demands → **overprovisioning**
- **heterogeneity** of hardware and software
- machine failures

Cloud Computing- Virtualization

-Virtualization is a basic enabler of Cloud Computing, it simplifies the management of physical resources for the three abstractions

For example, the state of a virtual machine (VM) running under a virtual machine monitor (VMM) can de saved and migrated to another server to balance the load

For example, virtualization allows users to operate in environments they are familiar with, rather than forcing them to specific ones

Cloud Computing- Virtualization

***"Virtualization**, in computing, refers to the act of creating a **virtual (rather than actual) version of something**, including but not limited to a virtual computer hardware platform, operating system (OS), storage device, or computer network resources."*

- Virtualization abstracts the underlying resources;
- simplifies their use;
- isolates users from one another; and
- supports replication which increases the elasticity of a system

Cloud Computing- Virtualization

- Cloud resource virtualization is important for:
  - Performance isolation
    - as we can dynamically assign and account for resources across different applications
  - System security:
    - as it allows isolation of services running on the same hardware
  - Performance and reliability:
    - as it allows applications to migrate from one platform to another
  - The development and management of services offered by a provider

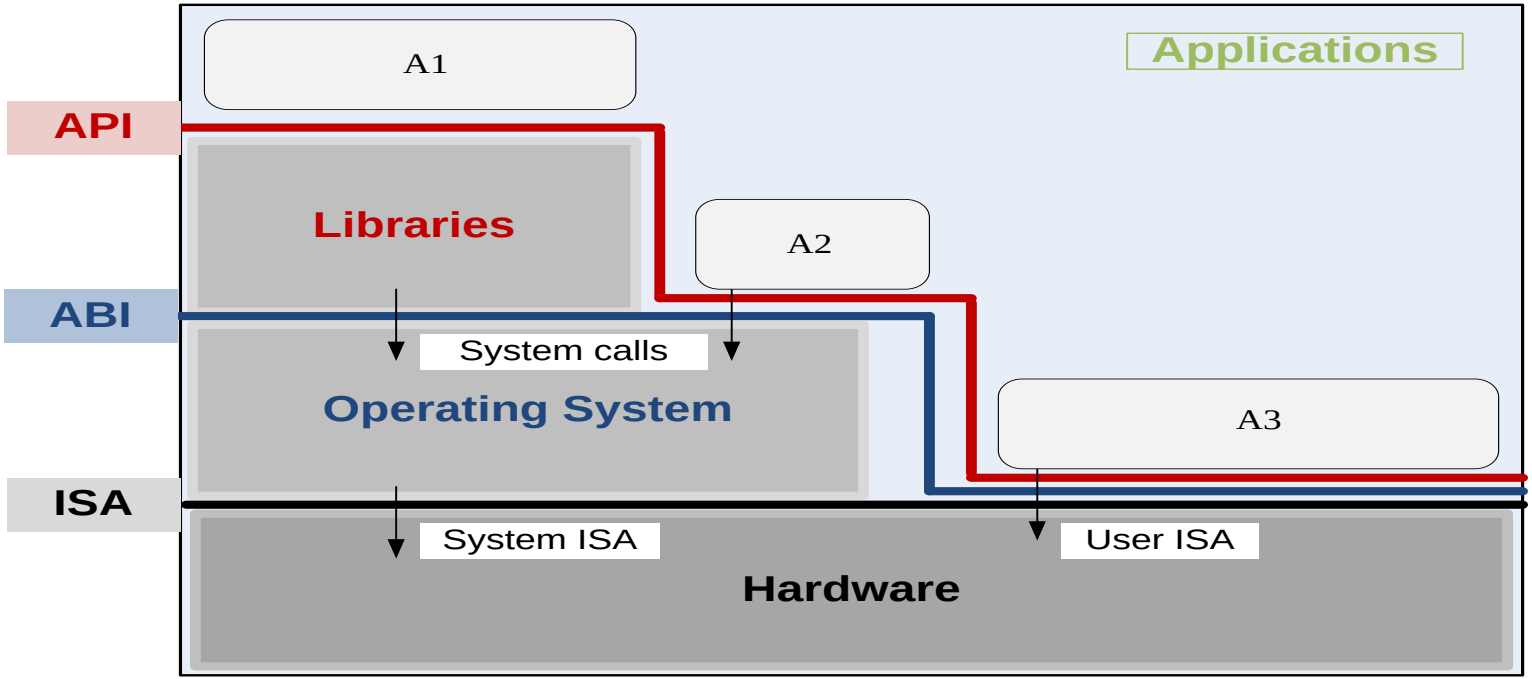
Cloud Computing- Virtualization

- Virtualization simulates the interface to a physical object by:
  - Multiplexing: creates multiple virtual objects from one instance of a physical object. Many virtual objects to one physical. Example - a processor is multiplexed among a number of processes or threads.
  - Aggregation: creates one virtual object from multiple physical objects. One virtual object to many physical objects. Example - a number of physical disks are aggregated into a RAID disk.
  - Emulation: constructs a virtual object of a certain type from a different type of a physical object. Example - a physical disk emulates a Random Access Memory (RAM).
  - Multiplexing and emulation. Examples - virtual memory with paging multiplexes real memory and disk; a virtual address emulates a real address.

Cloud Computing- Layering and Virtualization

- Layering – a common approach to manage system complexity:
  - Simplifies the description of the subsystems;
  - each subsystem is abstracted through its interfaces with the other subsystems
  - Minimizes the interactions among the subsystems of a complex system
  - With layering we are able to design, implement, and modify the individual subsystems independently
- Layering in a computer system:
  - Hardware
  - Software
    - Operating system
    - Libraries
    - Applications

Layering and Interfaces



Application Programming Interface (API), Application Binary Interface (ABI), and Instruction Set Architecture (ISA). An application uses library functions (A1), makes system calls (A2), and executes machine instructions (A3)

Interfaces

- **Instruction Set Architecture** (ISA) – at the boundary between hardware and software.

- **Application Binary Interface** (ABI) – allows the ensemble consisting of the application and the library modules to access the hardware; the ABI does not include *privileged* system instructions, instead it invokes system calls.

- **Application Program Interface** (API) - defines the set of instructions the hardware was designed to execute and gives the application access to the ISA; it includes high-level language (HLL) library calls which often invoke system calls

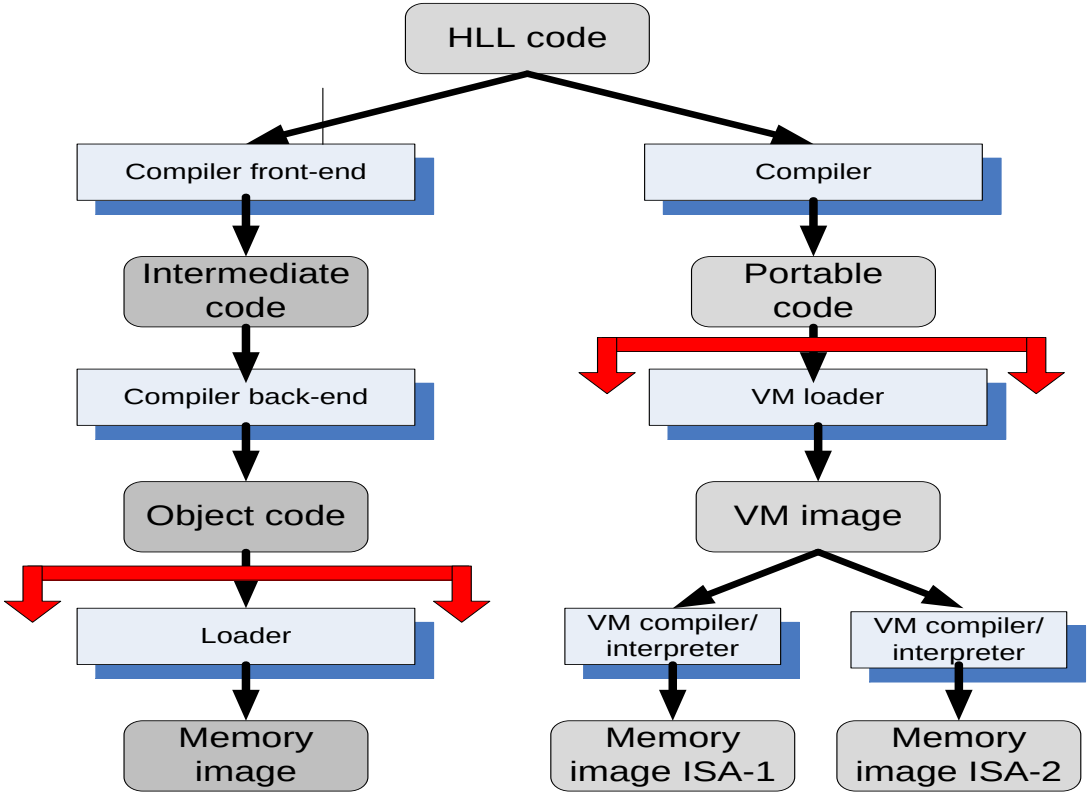
Code portability

- Binaries created by a compiler for a specific ISA and a specific operating systems are not portable

- It is possible, though, to compile a HLL program for a virtual machine (VM) environment where portable code is produced and distributed and then converted by binary translators to the ISA of the host system

- A **dynamic binary translation** converts blocks of guest instructions from the portable code to the host instruction and leads to a significant performance improvement, as such blocks are cached and reused

HLL Language Translations



History of Virtualization

(Reference "Modern Operating Systems" 4th Edition, p474 by Tanenbaum and Bos)

- **1960's, IBM: CP/CMS** control program: a virtual machine operating system for the IBM System/360 Model 67
- **2000, IBM: z-series** with 64-bit virtual address spaces and backward compatible with the System/360
- **1974: Popek and Golberg** from UCLA published *"Formal Requirements for Virtualizable Third Generation Architectures"* where they listed the conditions a computer architecture should satisfy to support virtualization efficiently. The popular x86 architecture that originated in the 1970s did not support these requirements for decades.
- **1990's, Stanford researchers, VMware:** Researchers developed a new hypervisor and founded VMware, the biggest virtualization company of today's. First virtualization solution was is 1999 for x86.
- Today many virtualization solutions: Xen from Cambridge, KVM, Hyper-V, …
- IBM was the first to produce and sell virtualization for the mainframe. But, VMware popularised virtualization for the masses.

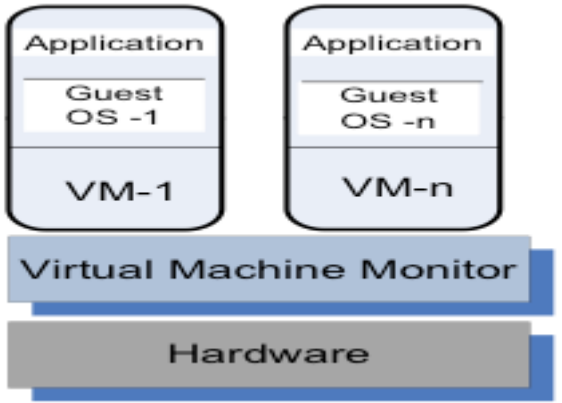
Virtual Machine Monitor (VMM / Hypervisor)

- A **virtual machine monitor (VMM/hypervisor)** partitions the resources of computer system into one or more **virtual machines (VMs)**. Allows several operating systems to run concurrently on a single hardware platform
- A VM is an execution environment that runs an OS
- VM – an isolated environment that appears to be a whole computer, but actually only has access to a portion of the computer resources
- A VMM allows:
  - Multiple services to share

    the same platform
  - Live migration - the movement

    of a server from one platform to another
  - System modification while maintaining

    backward compatibility with the original system
  - Enforces isolation among the systems, thus security



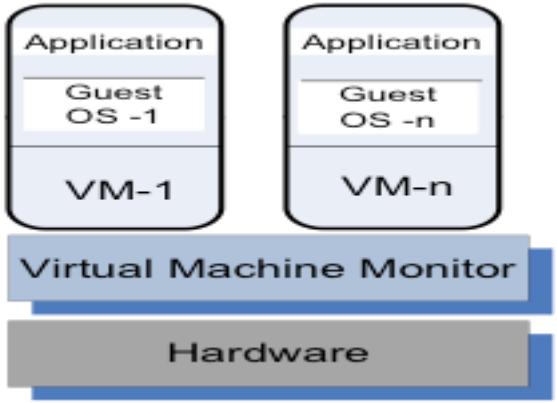
- A **guest operating system** is an OS that runs in a VM under the control of the VMM.

VMM Virtualizes the CPU and the Memory

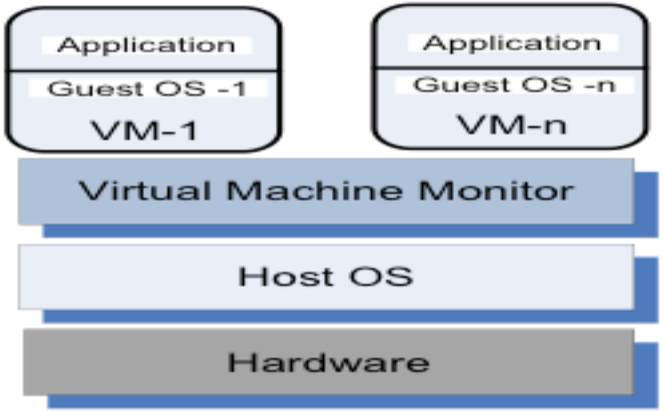
- A VMM (also hypervisor):
  - Traps the privileged instructions executed by a guest OS and enforces the correctness and safety of the operation
  - Traps interrupts and dispatches them to the individual guest operating systems
  - Controls the virtual memory management
  - Maintains a shadow page table for each guest OS and replicates any modification made by the guest OS in its own shadow page table. This shadow page table points to the actual page frame and it is used by the Memory Management Unit (MMU) for dynamic address translation.
  - Monitors the system performance and takes corrective actions to avoid performance degradation. For example, the VMM may swap out a VM to avoid thrashing.

Type 1 and 2 Hypervisors

Type 1 Hypervisor



Type 2 Hypervisor



- Taxonomy of VMMs:
  1. Type 1 Hypervisor (bare metal, native): supports multiple virtual machines and runs directly on the hardware (e.g., VMware ESX , Xen, Denali)
  2. Type 2 Hypervisor (hosted) VM - runs under a host operating system (e.g., user-mode Linux)

Examples of Hypervisors

Name	Host ISA	Guest ISA	Host OS	guest OS	Company
Integrity VM	*x86*-64	*x86*-64	HP-Unix	Linux,Windows HP Unix	HP
Power VM	Power	Power	No host OS	Linux, AIX	IBM
z/VM	z-ISA	z-ISA	No host OS	Linux on z-ISA	IBM
Lynx Secure	*x86*	*x86*	No host OS	Linux, Windows	LinuxWorks
Hyper-V Server	*x86*-64	*x86*-64	Windows	Windows	Microsoft
Oracle VM	*x86*, *x86*-64	*x86*, *x86*-64	No host OS	Linux, Windows	Oracle
RTS Hypervisor	*x86*	*x86*	No host OS	Linux, Windows	Real Time Systems
SUN xVM	*x86*, SPARC	same as host	No host OS	Linux, Windows	SUN
VMware EX Server	*x86*, *x86*-64	*x86*, *x86*-64	No host OS	Linux, Windows Solaris, FreeBSD	VMware
VMware Fusion	*x86*, *x86*-64	*x86*, *x86*-64	MAC OS *x86*	Linux, Windows Solaris, FreeBSD	VMware
VMware Server	*x86*, *x86*-64	*x86*, *x86*-64	Linux, Windows	Linux, Windows Solaris, FreeBSD	VMware
VMware Workstation	*x86*, *x86*-64	*x86*, *x86*-64	Linux, Windows	Linux, Windows Solaris, FreeBSD	VMware
VMware Player	*x86*, *x86*-64	*x86*, *x86*-64	Linux Windows	Linux, Windows Solaris, FreeBSD	VMware
Denali	*x86*	*x86*	Denali	ILVACO, NetBSD	University of Washington
Xen	*x86*, *x86*-64	*x86*, *x86*-64	Linux Solaris	Linux, Solaris NetBSD	University of Cambridge

Performance and Security Isolation

- The run-time behavior of an application is affected by other applications running concurrently on the same platform and competing for CPU cycles, cache, main memory, disk and network access. Thus, it is difficult to predict the completion time!
- Performance isolation - a critical condition for QoS guarantees in shared computing environments
- A VMM is a much simpler and better specified system than a traditional operating system. Example - Xen has approximately 60,000 lines of code; Denali has only about half: 30,000
- The security vulnerability of VMMs is considerably reduced as the systems expose a much smaller number of privileged functions. For example, Xen VMM has 28 hyper calls while Linux has 100s of system calls